

Technical Document #600: Database Systems - Comprehensive Overview

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ACID properties (Atomicity, Consistency, Isolation, Durability) ensure reliable transaction processing in databases.

Relational databases organize data into tables with rows and columns. SQL is used to query and manipulate structured data.

Sharding distributes data across multiple database servers. It improves performance by parallelizing query processing.

Database replication copies data across multiple servers. It improves availability and provides fault tolerance.

Computer vision applications are becoming ubiquitous, from facial recognition to autonomous driving. Deep learning has enabled breakthroughs in visual understanding.

Voice assistants like Alexa and Siri use speech recognition and natural language understanding. They have become integral parts of smart home ecosystems.

The Internet of Things (IoT) connects billions of devices worldwide. This connectivity generates massive amounts of data that can be analyzed for insights.

Digital twins are virtual replicas of physical systems. They enable simulation and optimization without risking real-world resources.

Key Takeaways:

- Database replication copies data across multiple servers.
- ACID properties (Atomicity, Consistency, Isolation, Durability) ensure reliable transaction processing in databases.
- Sharding distributes data across multiple database servers.